Composing scores beyond two levels: three-level, L-level and graph-structured hierarchies

COMPASS working note (derivation only, no code changes)

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Abstract

The hierarchical composition rule behind `gauss_hierarchical` and `gauss_jacobian` composes a single-observation score network over a two-level model (shared g, one local ℓ_j per observation). We show that the rule generalises without new approximations. The same single Gaussian-backward-kernel approximation gives (i) an explicit three-level rule, (ii) an L-level rule for any rooted tree, obtained by applying the two-level rule recursively ("a group is a big observation"), and (iii) a rule for any hierarchical model whose latent structure is a directed acyclic graph (DAG) and whose observations are sinks. In all three cases the tall-data backward kernel factorises *exactly* into single-observation kernels and prior-marginal kernels, weighted by integer *counting numbers*. The Gaussian step then reduces the composed score to a sparse linear system $\Lambda u = r$ assembled only from the network's own (conditional) scores and precisions. On trees, block elimination from the leaves up solves the system in time linear in the number of nodes, with no fill-in. On general DAGs the system is still exact, but its cost is set by the treewidth of a derived graph. The rule needs one new ingredient from three levels on: the diffused prior score and backward precision on every intermediate *path* (e.g. $p_t(g, h_k)$), not only on g. A NumPy check on linear-Gaussian trees (3 and 5 levels), a crossed-effects design and a non-tree DAG reproduces the exact diffused tall-data score to machine precision ($\leq 7 \times 10^{-15}$ relative error) at every noise level. Dropping the intermediate prior corrections gives 1–67% error.

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1 Recap: the two-level rule

The current model is $p(g, \ell_{1:n}, \mathbf{x}_{1:n}) = p(g) \prod_j p(\ell_j \mid g)\, p(x_j \mid g, \ell_j)$. The network provides, for each observation, the joint score $s_j = (s_j^g, s_j^\ell)$ of $p_t(\vartheta_j \mid x_j)$ at the noisy $y_j = (g_t, \ell_{j,t})$, with $\vartheta_j = (g, \ell_j)$. Let φ_λ be the isotropic Gaussian kernel at noise scale $\lambda = \lambda_t$ (VE convention, as in the paper). The derivation (paper, §3) has four steps:

1. **Exact factorisation:** $p(\theta \mid \mathbf{x}) \propto p(g)^{1-n} \prod_j p(\vartheta_j \mid x_j)$. Because the forward kernel factorises over coordinates, the same holds for the backward kernels: $\pi_t(\theta_0 \mid y, \mathbf{x}) \propto \pi_t^0(g_0 \mid g_t)^{1-n} \prod_j \pi_t(\vartheta_{j,0} \mid y_j, x_j)$.
2. **Tweedie:** the composed score is $u = (\mathbb{E}[\theta_0 \mid y, \mathbf{x}] - y)/\lambda^2$.
3. **Gaussian factors:** replacing each factor by a Gaussian with the correct (Tweedie) mean and precision P_j (resp. P_0) gives $\Lambda u = r$ with $\Lambda = (1-n)E_g^\top P_0 E_g + \sum_j E_j^\top P_j E_j$ and $r = (1-n)E_g^\top P_0 s_0 + \sum_j E_j^\top P_j s_j$. Λ is an arrowhead matrix.
4. **Elimination:** with $P_j = \left(\begin{smallmatrix} A_j & B_j \\ B_j^\top & C_j \end{smallmatrix}\right)$, $S_j = A_j - B_j C_j^{-1} B_j^\top$ and $R_j = -C_j^{-1} B_j^\top$,

$$u^g = \Big[\sum_j S_j + (1-n)P_0\Big]^{-1}\Big[\sum_j S_j s_j^g + (1-n)P_0 s_0\Big], \qquad u^{\ell_j} = s_j^\ell + R_j(u^g - s_j^g). \tag{1}$$

The rest of this note reuses steps 2 and 3 unchanged. All the work goes into (a) finding the right exact factorisation (step 1) for deeper or graph-structured models, and (b) solving the resulting larger sparse system (step 4).

Two observations that make everything below work. First, steps 2–3 never used the arrowhead structure. They hold for *any* finite product of Gaussian factors with real (possibly negative) exponents, provided every coordinate of y receives total noise exponent one (Proposition 4). Second, step 4 is a single instance of a generic operation, the elimination of a block that only one (merged) factor touches (Lemma 5). Applying it repeatedly gives the multi-level rules.

2 General setting and the exact factorisation

2.1 Model class

Assumption 1 (Hierarchical model on a DAG)**.** The latent variables are blocks $\theta_v \in \mathbb{R}^{d_v}$ indexed by the nodes $v \in V$ of a DAG G with parent sets $\mathrm{pa}(v)$, and

$$p(\theta, \mathbf{x}) = \prod_{v \in V} p(\theta_v \mid \theta_{\mathrm{pa}(v)}) \prod_{o \in O} p(x_o \mid \theta_{\mathrm{pa}(o)}),$$

where every observation $o \in O$ is a sink: $\mathrm{pa}(o) \subseteq V$, and no latent or other observation has o as a parent. Every latent node has at least one observed descendant (latent nodes without one are barren; integrate them out first, which leaves the model in the same class).

The two-level model is the DAG $g \to \ell_j \to x_j$ (with also $g \to x_j$). The three-level model is $g \to h_k \to \ell_{kj} \to x_{kj}$. Crossed random effects $g \to a_i$, $g \to b_j$, $(a_i, b_j) \to x_{ij}$ are a DAG that is not a tree.

Definition 1 (Observation regions)**.** For $o \in O$, let $R_o := \mathrm{An}(\mathrm{pa}(o))$ be the ancestral closure of its parents: all latent nodes from which there is a directed path into o, including $\mathrm{pa}(o)$. Write $\theta_R = (\theta_v)_{v \in R}$ and $d_R = \sum_{v \in R} d_v$.

Lemma 1 (Ancestral marginals are products of CPDs)**.** *For any ancestral set $R \subseteq V$ (closed under taking parents), $p(\theta_R) = \prod_{v \in R} p(\theta_v \mid \theta_{\mathrm{pa}(v)})$.*

Proof. Integrate out the nodes of $V \setminus R$ in reverse topological order. Each one integrated out is a sink of the current graph, since its children are outside R (otherwise R would not be ancestral) and have already been removed. Its CPD therefore integrates to one, and no remaining factor depends on it. □

The training unit of the network is consequently *one observation together with its ancestral region*: draw θ_{R_o} by ancestral sampling, then x_o. The network learns $p(\theta_{R_o} \mid x_o)$ under the correct prior marginal $p(\theta_{R_o})$, and no simulation ever involves more than one observation. In the two-level model $R_j = \{g, \ell_j\}$, which is the current single-group training unit. In the three-level model $R_{kj} = \{g, h_k, \ell_{kj}\}$, the root-to-leaf path.

2.2 Counting numbers

Definition 2 (Region family and counting numbers)**.** Let $\mathcal{R}$ be the closure of $\{R_o : o \in O\}$ under non-empty intersection. Intersections of ancestral sets are ancestral, so every $R \in \mathcal{R}$ is ancestral. Define the (Möbius) counting numbers top-down,

$$c(R) := 1 - \sum_{R' \in \mathcal{R},\, R' \supsetneq R} c(R'), \tag{2}$$

let $m(R) := |\{o : R_o = R\}|$ be the number of observations whose region is R, and define the *prior exponent* $\kappa(R) := c(R) - m(R)$.

Lemma 2 (Every node is covered exactly once)**.** *For every latent node v,*

$$\underbrace{|\{o \in O : v \in R_o\}|}_{\text{observation factors}} + \sum_{R \in \mathcal{R},\, R \ni v} \kappa(R) = \sum_{R \in \mathcal{R},\, R \ni v} c(R) = 1. \tag{3}$$

Proof. The first equality is the definition of κ. For the second, let $S_v := \bigcap \{R \in \mathcal{R} : v \in R\}$. The set is non-empty by Assumption 1, and the intersection contains v, so $S_v \in \mathcal{R}$. A region contains v if and only if it contains S_v. Hence $\sum_{R \ni v} c(R) = c(S_v) + \sum_{R \supsetneq S_v} c(R) = 1$ by (2). □

Theorem 3 (Exact tall-data factorisation on a DAG)**.** *Under Assumption 1,*

$$p(\theta \mid \mathbf{x}) \;\propto\; \prod_{o \in O} p(\theta_{R_o} \mid x_o) \prod_{R \in \mathcal{R}} p(\theta_R)^{\kappa(R)}, \tag{4}$$

and for every t the backward kernel of the diffused tall posterior factorises exactly as

$$\pi_t(\theta_0 \mid y, \mathbf{x}) \;\propto\; \prod_{o \in O} \pi_t(\theta_{R_o,0} \mid y_{R_o}, x_o) \prod_{R \in \mathcal{R}} \pi_t^0(\theta_{R,0} \mid y_R)^{\kappa(R)}, \tag{5}$$

where $\pi_t^0(\theta_{R,0} \mid y_R) \propto \varphi_\lambda(y_R - \theta_{R,0})\, p(\theta_{R,0})$ is the backward kernel of the prior marginal on R.

Proof. Clean posterior. By Bayes and Lemma 1, $p(\theta_{R_o} \mid x_o) \propto p(x_o \mid \theta_{\mathrm{pa}(o)}) \prod_{v \in R_o} p(\theta_v \mid \theta_{\mathrm{pa}(v)})$, and $p(\theta_R) = \prod_{v \in R} p(\theta_v \mid \theta_{\mathrm{pa}(v)})$. In the right-hand side of (4) each likelihood appears once. The CPD of node v appears with total exponent equal to the left-hand side of (3), which is 1. The right-hand side is therefore proportional to the joint $p(\theta, \mathbf{x})$, hence to $p(\theta \mid \mathbf{x})$.

Backward kernel. Since $\varphi_\lambda(y - \theta_0) = \prod_v \varphi_\lambda(y_v - \theta_{v,0})$ and, by (3), the exponents of the copies handed to the factors sum to one for each v,

$$\varphi_\lambda(y - \theta_0) = \prod_o \varphi_\lambda(y_{R_o} - \theta_{R_o,0}) \prod_R \varphi_\lambda(y_R - \theta_{R,0})^{\kappa(R)}.$$

Multiply this by (4) and pair each noise factor with its density factor. Each pair is a backward kernel times a normaliser ($p_t(y_{R_o} \mid x_o)$ or $p_t(y_R)^{\kappa(R)}$) that depends on y but not on θ_0. □

Remark 1. The counting condition (3) has the same form as the counting-number condition of region-based (Kikuchi) approximations in graphical models (Yedidia, Freeman & Weiss, 2005). Here, however, it is *exact*, because the regions are ancestral (Lemma 1). The Möbius choice (2) is canonical but not unique. Any integer κ satisfying (3) gives an exact factorisation; different solutions can differ in numerical conditioning once the factors are approximated.

2.3 From the factorisation to a linear system

Proposition 4 (Composed score as a sparse linear system)**.** *Index all factors in (5) by $f \in \mathcal{F}$. Each has weight w_f (1 for an observation factor, $\kappa(R)$ for a prior factor), scope R_f with selector E_f ($E_f\theta = \theta_{R_f}$), score $s_f = \nabla \log p_t(y_{R_f} \mid \cdot)$ and a Gaussian surrogate backward precision $P_f \succ 0$. If*

$$\Lambda := \sum_{f \in \mathcal{F}} w_f \, E_f^\top P_f E_f \;\succ 0, \qquad r := \sum_{f \in \mathcal{F}} w_f \, E_f^\top P_f s_f, \tag{6}$$

then the Gaussian surrogate of (5) gives the composed score $u = \nabla_y \log p_t(y \mid \mathbf{x}) \approx \Lambda^{-1} r$. Given exact factor scores and precisions, this is exact whenever all factors are Gaussian, in particular for every linear-Gaussian DAG.

Proof. Verbatim the paper's Step 3 with $(1 - n)$ and 1 replaced by w_f. With $m_f = y_{R_f} + \lambda^2 s_f$ (Tweedie), the log surrogate is $-\frac{1}{2}\theta_0^\top \Lambda \theta_0 + \theta_0^\top b$ with $b = \sum_f w_f E_f^\top P_f m_f$. Its mean is $\Lambda^{-1} b$, and $u = (\Lambda^{-1} b - y)/\lambda^2 = \Lambda^{-1}(b - \Lambda y)/\lambda^2$. Since $E_f y = y_{R_f}$, $b - \Lambda y = \sum_f w_f E_f^\top P_f (m_f - y_{R_f}) = \lambda^2 r$. □

The *sparsity graph* of Λ connects v and v' only if they share a region. It is therefore the union of the cliques $\{R_o\}$, which we call the *ancestral observation graph*. All structural results below concern how to solve $\Lambda u = r$ efficiently on this graph. The *values* of u do not depend on how it is solved.

2.4 The elimination lemma

Lemma 5 (Merge and marginalise)**.** ***(Merge.)*** *Factors with the same scope can be summed. Weighted factors (w_i, P_i, s_i) on one scope are equivalent in Λ, r to one factor (Q, q) with $Q = \sum_i w_i P_i$ and $q = \sum_i w_i P_i s_i$. If $Q \succ 0$, write $q = Q\sigma$ with the* merged score *$\sigma := Q^{-1} q$.*

(Marginalise.) *Let a block c of the unknowns appear in exactly one (merged) factor $(Q, Q\sigma)$, with scope (a, c) and $Q = \left(\begin{smallmatrix} Q_{aa} & Q_{ac} \\ Q_{ca} & Q_{cc} \end{smallmatrix}\right)$, $Q_{cc} \succ 0$. Replacing this factor by the factor $(S, S\sigma^a)$ on scope a alone, with*

$$S = Q_{aa} - Q_{ac} Q_{cc}^{-1} Q_{ca} = \left([Q^{-1}]_{aa}\right)^{-1}, \tag{7}$$

and back-substituting

$$u^c = \sigma^c + K\,(u^a - \sigma^a), \qquad K := -Q_{cc}^{-1} Q_{ca}, \tag{8}$$

leaves the solution u of $\Lambda u = r$ unchanged.

Proof. Merging holds because Λ and r are sums over factors. For marginalising, the c-rows of $\Lambda u = r$ read $Q_{ca}u^a + Q_{cc}u^c = Q_{ca}\sigma^a + Q_{cc}\sigma^c$, which gives (8). The factor's contribution to the a-rows is $Q_{aa}u^a + Q_{ac}u^c$ on the left and $Q_{aa}\sigma^a + Q_{ac}\sigma^c$ on the right. Substituting u^c cancels every σ^c term and leaves $Su^a = S\sigma^a$. Nothing else in the system involves c. □

This is exactly the paper's Step 4. In the two-level rule, $c = \ell_j$, $a = g$, the factor is (P_j, s_j), and the merged score is the network score s_j itself, so $S = S_j$, $K = R_j$ and the message is (S_j, s_j^g). The "cancellation of the local scores" is the statement that the message keeps σ^a, the *conditional* score of the factor at the current c. The marginal score of the scope a is never required. The lemma shows this cancellation happens at *every* elimination, at any depth and in any graph.

3 Three levels

3.1 Model and exact factorisation

$$p(\theta, \mathbf{x}) = p(g) \prod_{k=1}^{K} \Big[p(h_k \mid g) \prod_{j=1}^{n_k} p(\ell_{kj} \mid g, h_k)\, p(x_{kj} \mid g, h_k, \ell_{kj}) \Big],$$

with $g \in \mathbb{R}^{d_g}$ (global), $h_k \in \mathbb{R}^{d_h}$ (group), $\ell_{kj} \in \mathbb{R}^{d_\ell}$ (local) and $N = \sum_k n_k$ observations. Example: a population mean g, school effects h_k, pupil effects ℓ_{kj}, and pupil data x_{kj}. The network is trained on paths $\vartheta_{kj} = (g, h_k, \ell_{kj}) \in \mathbb{R}^D$, $D = d_g + d_h + d_\ell$, and returns $s_{kj} = (s^g_{kj}, s^h_{kj}, s^\ell_{kj})$ at $y_{kj} = (g_t, h_{k,t}, \ell_{kj,t})$.

The regions are $R_{kj} = \{g, h_k, \ell_{kj}\}$. Their intersections are $\{g, h_k\}$ (within a group) and $\{g\}$ (across groups). Definition 2 gives $c(R_{kj}) = 1$, $c(\{g, h_k\}) = 1 - n_k$ and $c(\{g\}) = 1 - \sum_k [n_k + (1 - n_k)] = 1 - K$. Hence

$$p(\theta \mid \mathbf{x}) \propto p(g)^{1-K} \prod_{k=1}^{K} p(g, h_k)^{1-n_k} \prod_{k,j} p(\vartheta_{kj} \mid x_{kj}). \tag{9}$$

A direct check: $p(g)$ appears with exponent $(1 - K) + \sum_k (1 - n_k) + \sum_k n_k = 1$, $p(h_k \mid g)$ with exponent $(1 - n_k) + n_k = 1$, and $p(\ell_{kj} \mid g, h_k)$ once. The same exponents distribute the noise kernel, so by Theorem 3

$$\pi_t(\theta_0 \mid y, \mathbf{x}) \propto \pi_t^0(g_0 \mid g_t)^{1-K} \prod_k \pi_t^0(g_0, h_{k,0} \mid g_t, h_{k,t})^{1-n_k} \prod_{k,j} \pi_t(\vartheta_{kj,0} \mid y_{kj}, x_{kj}).$$

What is new compared with two levels. The middle factor $p(g, h_k)^{1-n_k}$ is a correction on the *joint* prior marginal of the group path. Its diffused score $s_{0,k} = \nabla \log p_t(g_t, h_{k,t})$ and backward precision $P_{0,k} \in \mathbb{R}^{(d_g+d_h)^2}$ are needed. The conditional prior $p(h_k \mid g)$ alone is not enough, because diffusion acts on (g, h_k) jointly and $p_t(g, h_k) \neq p_t(g)\, p_t(h_k \mid g)$ for any useful sense of the latter. With two levels, the corresponding factor $p(g, \ell_j)^{1-1}$ has exponent zero, which is why it never showed up.

3.2 Linear system

Denote the top prior factor by (P_0, s_0) on g, the group prior factors by $(P_{0,k}, s_{0,k})$ on $a_k := (g, h_k)$, and the observation factors by (P_{kj}, s_{kj}) on ϑ_{kj}. Then (6) reads

$$\Lambda = (1 - K) E_g^\top P_0 E_g + \sum_k (1 - n_k) E_{a_k}^\top P_{0,k} E_{a_k} + \sum_{k,j} E_{kj}^\top P_{kj} E_{kj},$$

and r is the same with each P replaced by Ps. Ordered as $(g;\, h_1, \ell_{1\cdot};\, \ldots;\, h_K, \ell_{K\cdot})$, Λ is a *nested arrowhead*. The g-row couples to everything. Each h_k couples to g and to its own $\ell_{k\cdot}$. Each ℓ_{kj} couples only to g and h_k.

3.3 Elimination in three passes

Split each observation precision with respect to (a_k, ℓ_{kj}) as $P_{kj} = \begin{pmatrix} A_{kj} & B_{kj} \\ B_{kj}^\top & C_{kj} \end{pmatrix}$, with $A_{kj} \in \mathbb{R}^{(d_g+d_h)^2}$.

Pass 1: locals. ℓ_{kj} appears only in factor kj. Lemma 5 with $c = \ell_{kj}$, $a = a_k$ gives

$$S_{kj} = A_{kj} - B_{kj}C_{kj}^{-1}B_{kj}^\top = \left([\Sigma_{t,kj}]_{aa}\right)^{-1}, \quad K_{kj} = -C_{kj}^{-1}B_{kj}^\top, \quad u^{\ell_{kj}} = s_{kj}^{\ell} + K_{kj}(u^{a_k} - s_{kj}^{a}), \tag{10}$$

where $s_{kj}^a = (s_{kj}^g, s_{kj}^h)$. Factor kj becomes the message (S_{kj}, s_{kj}^a) on $a_k = (g, h_k)$.

Pass 2: groups. All remaining factors involving h_k now have scope a_k: the n_k messages and the group prior. Merge them:

$$Q_k = \sum_{j=1}^{n_k} S_{kj} + (1 - n_k)P_{0,k}, \qquad q_k = \sum_{j=1}^{n_k} S_{kj}s_{kj}^a + (1 - n_k)P_{0,k}s_{0,k}, \qquad \sigma_k = Q_k^{-1}q_k. \tag{11}$$

σ_k is exactly the two-level rule (1) applied to group k alone, with "global" $a_k = (g, h_k)$, locals $\ell_{k\cdot}$ and prior $p(g, h_k)$. It is the composed score of group k's tall posterior $p(g, h_k, \ell_{k\cdot} \mid x_{k\cdot})$ on its (g, h_k) block. Now split $Q_k = \begin{pmatrix} Q_k^{gg} & Q_k^{gh} \\ Q_k^{hg} & Q_k^{hh} \end{pmatrix}$ and eliminate h_k:

$$T_k = Q_k^{gg} - Q_k^{gh}(Q_k^{hh})^{-1}Q_k^{hg} = \left([Q_k^{-1}]_{gg}\right)^{-1}, \quad G_k = -(Q_k^{hh})^{-1}Q_k^{hg}, \quad u^{h_k} = \sigma_k^h + G_k(u^g - \sigma_k^g). \tag{12}$$

Group k becomes the message (T_k, σ_k^g) on g.

Pass 3: global.

$$u^g = \Big[\sum_{k=1}^{K} T_k + (1 - K)P_0\Big]^{-1}\Big[\sum_{k=1}^{K} T_k\,\sigma_k^g + (1 - K)P_0\,s_0(g_t)\Big]. \tag{13}$$

The three-level rule. Compute S_{kj}, K_{kj} (pass 1), then Q_k, σ_k, T_k, G_k (pass 2), then u^g (13). Back-substitute u^{h_k} by (12) and $u^{\ell_{kj}}$ by (10), using $u^{a_k} = (u^g, u^{h_k})$.

Read top-down, the global row (13) is the two-level rule with the *groups* as observations. Each group enters with its composed score σ_k^g and its effective precision T_k. Each group score is in turn the two-level rule over its own observations. The three-level rule is therefore the two-level rule nested in itself. The cost is N inversions of $d_\ell \times d_\ell$ blocks, K of $(d_g + d_h) \times (d_g + d_h)$ blocks, and one $d_g \times d_g$ solve: linear in N.

Remark 2 (Numerically preferred form). The merged score σ_k needs Q_k to be invertible. To eliminate h_k, only $Q_k^{hh} \succ 0$ is needed. Without forming σ_k, the message right-hand side is $\tau_k = q_k^g - Q_k^{gh}(Q_k^{hh})^{-1}q_k^h$ (which equals $T_k\sigma_k^g$ when Q_k is invertible), and the back-substitution is $u^{h_k} = (Q_k^{hh})^{-1}(q_k^h - Q_k^{hg}u^g)$. Implementations should use this information form.

3.4 Consistency checks

Each check follows because the rule returns $\Lambda^{-1}r$ and (Λ, r) coincide with the reference.

1. **One observation per group** ($n_k = 1$): the group priors have exponent 0, $Q_k = S_{k1}$ and $\sigma_k = s_{k1}^a$. The rule is the two-level rule with global g and local block (h_k, ℓ_{k1}).
2. **One group** ($K = 1$): the top prior has exponent 0 and $u^g = \sigma_1^g$. The rule is the two-level rule with global (g, h_1) and prior $p(g, h_1)$.
3. **No group parameters** ($d_h = 0$): $P_{0,k} = P_0$ and $T_k = Q_k$, so (13) becomes $\big[\sum_{kj} S_{kj} + (1 - N)P_0\big]^{-1}\big[\sum_{kj} S_{kj}s_{kj}^g + (1 - N)P_0 s_0\big]$, which is the two-level rule over all N observations.

4. $N = 1$**:** every prior exponent vanishes, and the rule returns the network score.

5. **Linear-Gaussian model:** exact at every t (Proposition 4); verified in Section 6.

4 L levels and arbitrary trees

4.1 Setting

Let the latent DAG be a rooted tree: every v has at most one parent, and the root ρ has none. Observations may attach to any node, provided all parents of an observation lie on one root path. Write $O(v)$ for the observations whose deepest parent is v (so $R_o = \pi(v)$), $\mathrm{ch}(v)$ for the children of v and $\mathrm{sub}(v)$ for the subtree rooted at v. Let $\pi(v)$ be the path from ρ to v (inclusive) and $\alpha(v) := \pi(v) \setminus \{v\}$ the strict ancestors, so that $\alpha(w) = \pi(v)$ for every $w \in \mathrm{ch}(v)$. This covers L-level nested designs, ragged trees, and observations on intermediate levels.

Proposition 6 (Counting numbers on a tree)**.** *On a tree, $\mathcal{R} = \{\pi(v)\}$, and the prior exponent of $\pi(v)$ is*

$$\kappa(v) := 1 - |\,\mathrm{ch}(v)| - |O(v)|. \tag{14}$$

Proof. Intersections of root paths are root paths. Given (14), check (3) by induction from the leaves: let $N(v) := \sum_{u \in \mathrm{sub}(v)} |O(u)|$ be the number of observations below v. Then

$$N(v) + \sum_{u \in \mathrm{sub}(v)} \kappa(u) = |O(v)| + \kappa(v) + \sum_{w \in \mathrm{ch}(v)} \Big[N(w) + \sum_{u \in \mathrm{sub}(w)} \kappa(u) \Big] = |O(v)| + \kappa(v) + |\,\mathrm{ch}(v)| = 1. \tag{15}$$

The left-hand side of (15) is exactly the coverage of v, and also of every ancestor of v, contributed by $\mathrm{sub}(v)$. Applied at the root, it shows every node is covered once. The Möbius numbers satisfy the same recursion, so they coincide with (14). Nodes with $\kappa(v) = 0$ (e.g. a single child and no observations) do not need a prior factor. □

For two levels, $\kappa(g) = 1 - n$ and $\kappa(\ell_j) = 0$. For three levels, $\kappa(g) = 1 - K$, $\kappa(h_k) = 1 - n_k$ and $\kappa(\ell_{kj}) = 0$.

4.2 The recursive rule

Theorem 7 (L-level / tree composition rule)**.** *Process the nodes in post-order (children before parents). For each v, all quantities live on the path $\pi(v)$:*

$$Q_v = \sum_{o \in O(v)} P_o + \kappa(v)\, P_{0,v} + \sum_{w \in \mathrm{ch}(v)} S_w, \tag{16}$$

$$\sigma_v = Q_v^{-1} \Big[\sum_{o \in O(v)} P_o s_o + \kappa(v)\, P_{0,v} s_{0,v} + \sum_{w \in \mathrm{ch}(v)} S_w\, \sigma_w^{\alpha(w)} \Big], \tag{17}$$

$$S_v = [Q_v]_{\alpha\alpha} - [Q_v]_{\alpha v} [Q_v]_{vv}^{-1} [Q_v]_{v\alpha}, \qquad K_v = -[Q_v]_{vv}^{-1} [Q_v]_{v\alpha} \qquad (\alpha = \alpha(v)), \tag{18}$$

where (P_o, s_o) are the network's precision and score for observation o on $\pi(v)$, and $(P_{0,v}, s_{0,v})$ are the prior backward precision and diffused prior score on $\pi(v)$. Then process the nodes top-down:

$$u^\rho = \sigma_\rho, \qquad u^v = \sigma_v^v + K_v (u^{\alpha(v)} - \sigma_v^{\alpha(v)}). \tag{19}$$

If every $[Q_v]_{vv} \succ 0$ and $Q_\rho \succ 0$, this u is the unique solution of $\Lambda u = r$ with the counting numbers (14). In particular, it is the exact diffused tall-data score of every linear-Gaussian tree.

Proof. By induction over the post-order, with the invariant: *when v is processed, all of* $\mathrm{sub}(v) \setminus \{v\}$ *has been eliminated, and the factors of the reduced system that involve block v are exactly the terms in (16), all with scope $\pi(v)$.* Any original factor that contains v has scope $\pi(u)$ with $u \in \mathrm{sub}(v)$.

Those with $u \neq v$ have been absorbed by the induction hypothesis into the messages $(S_w, \sigma_w^{\alpha(w)})$ of the children w, whose scope is $\alpha(w) = \pi(v)$. Those with $u = v$ are the observation factors $O(v)$ and the prior factor of v. Merging them (Lemma 5) gives $(Q_v, Q_v\sigma_v)$, the only factor containing v. Marginalising v (Lemma 5, $c = v$, $a = \alpha(v)$) produces the message $(S_v, \sigma_v^{\alpha(v)})$ on $\pi(\mathrm{pa}(v))$ and the back-substitution (19) (an instance of (8)), which maintains the invariant for $\mathrm{pa}(v)$. At the root, $Q_\rho u^\rho = Q_\rho \sigma_\rho$. Uniqueness: the elimination is a block $LDL^\top$ factorisation of Λ with pivots $[Q_v]_{vv}$ and Q_ρ, so $\Lambda \succ 0$ if and only if all pivots are positive definite. □

Interpretation: a subtree is a big observation. By (15), the factors inside $\mathrm{sub}(w)$ cover $\pi(\mathrm{pa}(w))$ exactly once as well. They therefore form the exact tall posterior of the *sub-model* consisting of the path to w, the subtree $\mathrm{sub}(w)$, and its observations. The message $(S_w, \sigma_w^{\alpha(w)})$ is the Gaussian summary of that sub-model's composed backward kernel on the parent path: S_w is its marginal precision on $\pi(\mathrm{pa}(w))$, and $\sigma_w^{\alpha(w)}$ is its composed, conditional score there. The parent combines its children exactly as the two-level rule combines observations. This is Gaussian belief propagation, in precision–score parameterisation, on the tree, run upwards once. The downward pass is back-substitution, not a second message pass, because only the mean (Tweedie) is needed.

Cost. $\sum_v \left(|O(v)| + |\,\mathrm{ch}(v)|\right) d_{\pi(v)}^2 + d_{\pi(v)}^3$: linear in the number of nodes and observations, and cubic in the path dimension $d_{\pi(v)} \leq \sum_{\text{levels}} d_{\text{level}}$. Λ has zero fill-in because its sparsity graph (the union of root-path cliques) is chordal, and the leaves-first order is a perfect elimination ordering.

4.3 Positive definiteness and the information form

Using $\kappa(v) = 1 - |O(v)| - |\,\mathrm{ch}(v)|$,

$$Q_v = P_{0,v} + \sum_{o \in O(v)} (P_o - P_{0,v}) + \sum_{w \in \mathrm{ch}(v)} (S_w - P_{0,v}) =: P_{0,v} + \sum_o I_o + \sum_w I_w. \tag{20}$$

This is the paper's information form, $\Lambda_S = P_0 + \sum_j I_j$, at every level. If every increment is positive semidefinite, then $Q_v \succeq P_{0,v} \succ 0$, all pivots are positive definite, and the rule is well posed. The increments are PSD in the exact linear-Gaussian case. I_o is PSD because P_o is the precision of $p(\theta_{\pi(v),0} \mid y_{\pi(v)}, x_o)$ and $P_{0,v}$ that of $p(\theta_{\pi(v),0} \mid y_{\pi(v)})$. I_w is PSD because S_w is the $\pi(v)$-marginal precision of the sub-model kernel, which additionally conditions on $y_{\mathrm{sub}(w)}$ and $x_{\mathrm{sub}(w)}$. Gaussian conditioning never increases covariance. The current PSD projection of I_j therefore extends directly: project every I_o and I_w at every level.

4.4 What each method has to supply

- **Observation factors.** The network must output the joint score on the whole path $\pi(v)$ for observation o. `gauss_hierarchical` needs a pilot covariance $\Sigma_{0,o}$ of $p(\theta_{\pi(v)} \mid x_o)$ (from the network's own draws), and sets $P_o = \Sigma_{0,o}^{-1} + \lambda^{-2}\mathbf{I}$. `gauss_jacobian` takes $P_o = \left[\lambda^2(\mathbf{I} + \lambda^2 \nabla s_o)\right]^{-1}$ from $d_{\pi(v)}$ JVPs or finite-difference copies. Both are unchanged; only the dimension grows.

- **Prior factors.** For every node with $\kappa(v) \neq 0$, the diffused prior score $s_{0,v}$ and backward precision $P_{0,v}$ on the *whole path* $\pi(v)$ are needed. With two levels, only $\pi(\rho) = \{g\}$ is needed, and a Gaussian $p(g)$ makes it closed-form. From three levels on, $p(g, h_k)$ appears. In partial-pooling models g contains the scale of h_k, so this prior is a funnel, not a Gaussian, and has no closed-form diffused score. The options are: (a) a prior-score network trained on root-path prior draws (cheap: no simulator calls), or the same network evaluated with all data masked if its training masks include the "no data" pattern; (b) its Tweedie Jacobian for $P_{0,v}$, as in `gauss_jacobian`; (c) a Gaussian moment surrogate from prior samples. Option (c) introduces an error that the two-level rule does not have when $p(g)$ is Gaussian. The

prior factors carry weights $1 - n_k$, so their errors are amplified by the group size, exactly as the top-level prior error is amplified by $n - 1$ today.

5 General DAG hierarchies

Theorem 3 and Proposition 4 already cover every model satisfying Assumption 1. Exactness of the factorisation therefore holds for every DAG hierarchy, with no running-intersection or tree assumption, because the regions are ancestral. What changes is the cost of solving $\Lambda u = r$.

Example 1 (Crossed random effects)**.** $g \to a_i$ $(i \le I)$, $g \to b_j$ $(j \le J)$, $x_{ij} \mid g, a_i, b_j$, one observation per cell. The regions are $\{g, a_i, b_j\}$, with intersections $\{g, a_i\}$, $\{g, b_j\}$ and $\{g\}$. The Möbius numbers are $c(\{g, a_i, b_j\}) = 1$, $c(\{g, a_i\}) = 1 - J$, $c(\{g, b_j\}) = 1 - I$, and

$$c(\{g\}) = 1 - \big[IJ + I(1 - J) + J(1 - I)\big] = (I - 1)(J - 1) \ \ge 0,$$

so

$$p(\theta \mid \mathbf{x}) \propto p(g)^{(I-1)(J-1)} \prod_i p(g, a_i)^{1-J} \prod_j p(g, b_j)^{1-I} \prod_{i,j} p(g, a_i, b_j \mid x_{ij}).$$

The global prior exponent is *positive*: the row and column corrections remove $p(g)$ too often, and it has to be multiplied back.

Example 2 (Shared child)**.** $g \to h_1, h_2$; $z_1 \mid h_1, h_2$; $z_2 \mid h_2$; observations $x_a, x_b \mid z_1$, $x_c \mid z_2$, $x_d \mid h_1$. The regions are $\{g, h_1, h_2, z_1\}$ (twice), $\{g, h_2, z_2\}$ and $\{g, h_1\}$. The prior factors are $p(g, h_1, h_2, z_1)^{-1}$, $p(g, h_1)^{-1}$ and $p(g, h_2)^{-1}$, and $\kappa(\{g\}) = 0$.

5.1 Solving the system

1. **Junction-tree case.** If the maximal regions can be arranged in a tree with the running-intersection property, i.e. the ancestral observation graph is chordal with this clique tree, Theorem 7 carries over with nodes replaced by cliques and blocks by clique-minus-separator sets. Each clique merges its factors, eliminates its private variables by Lemma 5, and passes a Schur message on the separator. The cost is linear in the number of cliques and cubic in the clique dimension. Rooted trees are the special case where the cliques are root paths.

2. **General case.** Pick an elimination order (e.g. minimum degree) and run sparse block Cholesky. Every elimination step is still Lemma 5, so only network conditional scores ever enter, but fill-in appears. The cost is governed by the treewidth of the ancestral observation graph. In Example 1, after eliminating the a_i, the b_j form a dense $J \times J$ block: $O(IJ^2 + J^3)$, fine for hundreds of levels.

3. **Large loopy designs.** Under the information-form condition $\Lambda \succ 0$, conjugate gradients needs only products $\Lambda v = \sum_f w_f E_f^\top P_f (E_f v)$, at cost $O(\sum_f d_{R_f}^2)$ per iteration. Alternatively, Gaussian BP on Λ returns the *exact* mean whenever it converges (Weiss & Freeman, 2001), which is all Tweedie needs. Convergence is guaranteed for walk-summable Λ.

5.2 Limits of the graph generalisation

1. **Bounded ancestral regions.** The training unit is (θ_{R_o}, x_o). If $|R_o|$ grows with the data, as in a latent Markov chain $z_1 \to z_2 \to \cdots$ with $x_t \mid z_t$ where $R_t = \{z_1, \ldots, z_t\}$, a single-observation network cannot be trained. Two exact alternatives exist:

   (a) *Decomposable prior over non-ancestral regions.* If $p(\theta) = \prod_C p(\theta_C) / \prod_S p(\theta_S)$ over a junction tree (e.g. $C_t = \{z_{t-1}, z_t\}$, $S_t = \{z_t\}$), and the network for x_o is trained on $(\theta_{C(o)}, x_o)$ with $\theta_{C(o)}$ drawn from the *true prior marginal* $p(\theta_{C(o)})$ (e.g. cropped windows of long simulations), the proof of Theorem 3 goes through with these regions. The counting condition is the junction-tree identity.

(b) *Training-prior ratio form.* For any regions $R_o \supseteq \mathrm{pa}(o)$ and any training prior $\tilde{p}_o(\theta_{R_o})$, $p(\theta \mid \mathbf{x}) \propto p(\theta) \prod_o \tilde{p}_o(\theta_{R_o} \mid x_o)/\tilde{p}_o(\theta_{R_o})$. Each coordinate is covered $1 + \sum_o (1-1) = 1$ times, so Theorem 3 and Proposition 4 hold with $\Lambda = P_0^{\text{full}} + \sum_o E_o^\top (P_o - \tilde{P}_{0,o}) E_o$. This needs the diffused prior score and precision on all of θ, which are tractable for Gaussian Markov priors (sparse P_0^{full}) but not in general.

2. **Observations must be sinks.** Models in which data feed back into latents or other data (autoregressive observations) are outside Assumption 1.

3. **The single approximation is unchanged, but it is applied in higher dimension.** Each observation factor is a Gaussian surrogate on d_{R_o} dimensions instead of $d_g + d_\ell$. The multimodality and heavy-tail failure modes documented for the two-level rule (e.g. the IK branch bias summed n times) apply per factor. Prior factors appear at every intermediate level with weights $1 - |O(v)| - |\mathrm{ch}(v)|$, so prior-score and Gaussian-surrogate errors are amplified at each level, not only at the root.

6 Numerical verification

A standalone NumPy script (CPU only, 3 threads; not part of the package) builds random linear-Gaussian DAGs, samples data, and computes the exact Gaussian posterior of every factor under the true prior marginal of its region, together with the exact diffused tall-data score $-(C + \lambda^2 \mathbf{I})^{-1}(y - m)$. It then compares (i) the dense solve $\Lambda^{-1} r$ with Möbius counting numbers, (ii) the recursive tree rule of Theorem 7 (trees only), and (iii) an ablation that keeps only the root prior correction and drops all intermediate prior factors. It also asserts (3) for every node. The table reports relative error $\|u - u_{\text{exact}}\|/\|u_{\text{exact}}\|$, maximised over $\lambda \in \{0.01, 0.3, 1, 5\}$ with a fresh random y per λ.

model	dim	obs	dense	tree rule	drop mid priors	non-zero κ
3-level tree ($K = 3$, $n_k = 2, 3, 4$, obs on h_0, 2 trials on ℓ_{21})	23	11	2.2×10^{-15}	2.3×10^{-15}	4.8% – 48%	g:−2, h_k:−2, −2, −3, ℓ_{21}:−1
5-level random tree	46	13	6.7×10^{-15}	6.7×10^{-15}	3.3% – 21%	11 path priors
crossed 3×4 (Example 1)	9	12	2.4×10^{-15}	–	8.6% – 67%	g:+6, a_i:−3, b_j:−2
shared child (Example 2)	5	4	4.7×10^{-16}	–	0.8% – 12%	three −1

The derived counting numbers are therefore exact on all four structures, and the recursive tree rule agrees with the dense solve to round-off. The ablation error is largest at small λ: the intermediate prior corrections are not optional, and are not a high-noise effect. The crossed design also confirms the positive global exponent $\kappa(g) = +6 = (3-1)(4-1)$.

7 Summary

1. **Three levels: yes.** Equations (10) to (13). The new required inputs are the network's path score and precision on (g, h_k, ℓ_{kj}), and the diffused prior score and precision on (g, h_k), which enter with weight $1 - n_k$.

2. **L levels / any rooted tree: yes.** Theorem 7 is the two-level rule applied recursively, with subtrees acting as observations. It has one post-order pass and one back-substitution, cost linear in the number of nodes, and an information form with a per-level PSD safeguard. It is exact on linear-Gaussian trees.

3. **Any DAG hierarchy with sink observations: yes for exactness, with structure-dependent cost.** The counting-number factorisation (Theorem 3) is exact for every DAG

because the regions are ancestral. The composed score is always $\Lambda^{-1}r$ built from network conditional scores. Junction-tree structures keep a closed-form message rule; loopy designs need a sparse or iterative solve.

4. **Main practical obstacles:** (i) diffused prior scores on intermediate paths (non-Gaussian funnels in partial pooling); (ii) the network input dimension grows with depth; (iii) unbounded ancestral sets (state-space models) need the non-ancestral or ratio variants of Section 5.2.